

# Signed covariance correction of a compact pulse induced by the coupled viscous pair

## 1 Source objects and the finite calculation

This calculation uses the preserved 165-page supplied manuscript *Finite Time Blowup for Navier–Stokes*, source equations (7.23)–(7.42), (9.12)–(9.14). Its PDF SHA-256 is

8c8a94ad9ac824c8b605b9827cadf7beaca48bd10b380de3cfc872a2c37afa81.

The source pulse fields, auxiliary rectangles, phases, and leading covariance columns are the given objects; their global construction is source input. Every finite map asserted below is proved here. The existing coupled-stage proofs remain in the complete 102-page reader. The companion pulse note proves the exact frame residual and a constructed finite-interval intertwiner along its evolving coefficients. The source uses positive  $q(z, t)$ , band scales  $Q_\beta = 2^{-\ell_\beta}$ ,  $\varepsilon_\beta = Q_\beta^h$ ,  $A = 1/2 + h$ ; none is set to one here or identified with the original coupled  $A_0, \lambda_0, \mu, \nu = \sqrt{A_0}$  or its physical diffusivities.

All operations in this note are on a compact set inside  $q > 0$  and the source active annulus. There are finitely many relevant bands and locally finite source labels. Supports of newly prescribed inputs are strictly inside the annulus, so no lower bound on a leading amplitude at a vanishing shell edge is assumed. Endpoint and shell-edge estimates are not inferred from compact-interior estimates.

## 2 Physical curl of the actual retained amplitude

Let  $y = (\Theta, \Omega)^T$  be the actual smooth retained coupled solution,  $y' = q_v(C_c y + c_c)$ , with the coefficients of the companion pulse note. Use the smooth actual coefficients on the chosen slow/auxiliary patch; all harmonic coefficients and cutoffs are independent of  $\theta$  in cylindrical components, and each carrier has nonzero integer angular frequency. For its source tangent frame  $B$ , smooth coefficient map  $L$ , and a fixed smooth real compact cutoff  $\chi$  of this type, put

$$t_m = \chi B L y, \quad \mathcal{E}_\chi = \chi \{ [L' + q_v L C_c - H L + m^2 d_N L] y + q_v L c_c \} + \chi' L y, \quad H = B_\ell (A_\Phi B - B').$$

Here prime differentiates along the pulse coordinate; the other slow and auxiliary derivatives remain in the spatial formulas below. Smooth parameter dependence of the companion's constructed  $L$  follows directly from its integral equations on a fixed interval: for a nonzero parameter multi-index  $I$ ,

$$\partial^I X(v) = \int_{v_0}^v \left[ D \partial^I X + \sum_{0 < J \leq I} \binom{I}{J} (\partial^J D) \partial^{I-J} X \right] dw.$$

The initial parameter derivatives vanish for identity initial data. Inductively the known forcing is smooth and bounded on the compact patch; the same convergent ordered-integral iteration solves

this equation. This proves every finite mixed derivative of  $X, X^{-1}, L$ . Since  $n^T B = 0$ ,  $n^T t_m = 0$ . Differentiation proves  $t'_m - \mathbf{A}_\Phi t_m + m^2 d_N t_m = B \mathcal{E}_\chi$ . Thus even the pulse-cutoff term is explicit.

A physical version of the exact curl construction makes the intermediate map to momentum residual unambiguous. For each chosen harmonic let  $\phi(x, t) = k_\beta m \Phi_\beta(x, t, Y(x, t))$  be its real evaluated phase, let  $\xi = \nabla_x \phi \neq 0$ , and let  $a(x, t) = Q_\beta^{-A} t_m$  be the physical transverse amplitude. The source coordinate change is  $r = \sqrt{Q_\beta} R$ ,  $z = Q_\beta^D Z$ ,  $1 - t = Q_\beta T$ , with  $D = 1/2 - h$  and normalized axial derivative  $D_z = \varepsilon_\beta \partial_Z$ . It gives  $\xi = (k_\beta m / \sqrt{Q_\beta}) n_\Phi$ ; all auxiliary chain derivatives are included in  $n_\Phi$ . Hence  $\xi \cdot a = 0$ . Define in Cartesian components

$$C = \frac{i \xi \times a}{|\xi|^2}, \quad \mathcal{A} = C e^{i\phi}, \quad \nabla \times \mathcal{A} = (a + r_a) e^{i\phi}, \quad r_a = \nabla \times C. \quad (1)$$

The vector triple-product identity gives  $i \xi \times C = a$ , proving the last equality. It also shows that the physical potential has exactly the source factor  $Q_\beta^{1/2-A}$  relative to its chart coefficient  $i n_\Phi \times t_m / (k_\beta m |n_\Phi|^2)$ . Cartesian differentiation of  $C$  includes derivatives of the cylindrical basis; the same terms are the  $R^{-1}$  terms of source (7.37)–(7.38). Take real parts and sum the finite retained harmonics. This constructs a smooth compact velocity  $w_y$  with  $\nabla \cdot w_y = 0$ .

Retain a finite smooth source state  $(u_*, p_*)$ , including its background, actual leading waves and their curl corrections, and the real physical pressure increments obtained from the companion pulse pressure formula with  $f_m = 0$ :

$$\pi_{\chi, m} = \frac{i}{k_\beta m |n|^2} (n^T \mathbf{K} t_m - (n')^T t_m), \quad \pi_y = \sum_{\beta, m} \text{Re}(Q_\beta^{-2A} \pi_{\chi, m} e^{i\phi_{\beta, m}}).$$

These are smooth and compact wherever  $t_m$  is. This definition fixes the pressure map without assuming the pulse residual vanishes. Define the actual candidate residual

$$R_y = R_{\nu_{\text{NS}}}(u_* + w_y, p_* + \pi_y), \\ R_{\nu_{\text{NS}}}(u, p) := \partial_t u + (u \cdot \nabla) u - \nu_{\text{NS}} \Delta u + \nabla p. \quad (2)$$

This is a definition by explicit differentiated fields, not an assumed source term. Its exact expansion is

$$R_y = R_{\nu_{\text{NS}}}(u_*, p_*) + \partial_t w_y + (u_* \cdot \nabla) w_y + (w_y \cdot \nabla) u_* - \nu_{\text{NS}} \Delta w_y + \nabla \pi_y + (w_y \cdot \nabla) w_y.$$

Every cutoff, parent, phase, and curl derivative occurs in this formula. Physical  $\nu_{\text{NS}}$  is retained and is distinct from every source time-scaling convention. Define  $\Delta R_y = R_y - R_{\nu_{\text{NS}}}(u_*, p_*)$ . We correct the new compact increment; this difference has compact support since every added term does. The baseline residual remains explicitly.

The averaging needed next is the source extended-field operation, not a spatial average of the evaluated field. Evaluate derivatives on the extended coefficients first by the complete chain rule. In a fixed chart set  $\mathcal{D}_i = \partial_{x_i} + \sum_\alpha (\partial_{x_i} Y_\alpha) \partial_{Y_\alpha}$  and similarly  $\mathcal{D}_t$ . These are the pullbacks of commuting physical derivatives, so  $[\mathcal{D}_i, \mathcal{D}_j] = 0$  on their lifted domain. For smooth periodic coefficients their auxiliary derivative terms have zero Haar average. In cylindrical components angular differentiation also differentiates the basis. This defines an extended residual  $\tilde{R}_y$  whose evaluation is exactly (2). Average the tangential components of the increment at fixed slow variables:

$$F_2 = \langle \Delta \tilde{R}_{y, \theta} \rangle, \quad F_1 = \langle \Delta \tilde{R}_{y, z} \rangle, \quad \langle g \rangle = \int_{\mathbb{T}^2} \frac{1}{2\pi} \int_0^{2\pi} g \, d\theta \, dY. \quad (3)$$

These  $F_e$  are smooth and compact in the chosen physical annulus. Equations (1)–(3) are the explicit intermediate map from the retained coupled pair to a signed momentum residual. They do not identify it with the original two-dimensional Boussinesq force.

### 3 Radial moments, support, and the original chart scales

For each fixed  $(z, t)$ , choose  $0 < r_- < r_+$  containing the compact supports of both  $F_e$  and the chosen bumps in the active annulus. The source bump is

$$b_e(r, z, t) = q^{-(e+1)/2} \widehat{b}_e(r/\sqrt{q}), \quad \int_0^\infty x^e \widehat{b}_e(x) dx = 1, \quad e = 1, 2,$$

with interior support. The change  $r = \sqrt{q}x$  proves  $\int r^e b_e dr = 1$  exactly. Define

$$M_e = \int_0^\infty r^e F_e(r) dr, \quad \sigma_e(r) = -r^{-e} \int_0^r s^e (F_e(s) - b_e(s) M_e) ds. \quad (4)$$

Then

$$(\partial_r + e/r)\sigma_e = -F_e + b_e M_e, \quad \int_0^\infty r^e (F_e - b_e M_e) dr = 0. \quad (5)$$

Differentiate  $r^e \sigma_e$  to prove the first identity; integrate and use the bump normalization for the second. The integral is zero below  $r_-$  and above  $r_+$ , so  $\sigma_e$  has compact smooth support in the same enclosing interval. A homogeneous solution is  $cr^{-e}$ , and a compact homogeneous solution has  $c = 0$ ; thus the primitive is unique among compact solutions of its displayed equation. Moreover a compact solution of  $(\partial_r + e/r)\sigma = -F_e$  exists if and only if  $M_e = 0$ : necessity follows by integrating the weighted derivative, and sufficiency follows from (4). The two moments are therefore the exact surviving finite-dimensional terms.

For  $C_e = \int_{r_-}^{r_+} s^e ds$ ,  $|M_e| \leq C_e \|F_e\|_\infty$  and

$$\|\sigma_e\|_\infty \leq r_-^{-e} C_e (\|F_e\|_\infty + \|b_e\|_\infty |M_e|).$$

If  $G_e = F_e - b_e M_e$ , successive radial derivatives obey the exact recurrence

$$\partial_r^{j+1} \sigma_e = -\partial_r^j G_e - e \sum_{i=0}^j \binom{j}{i} (-1)^i i! r^{-i-1} \partial_r^{j-i} \sigma_e.$$

This proves every finite radial derivative bound inductively. Parameter derivatives follow by differentiating the compact integral; all derivatives of  $q, b_e$  and the moving physical data are retained. On the present compact set  $q$  is bounded away from zero. No terminal uniform estimate is implied.

The source (9.12)–(9.14) uses physical  $F_e = Q^{-2A-1/2} E_e^*$ , where  $E_2^* = \langle E_\theta \rangle_Y$  and  $E_1^* = \langle E_z \rangle_Y$ . For our fields define  $E_e^* = Q^{2A+1/2} F_e$  in that same convention. With  $r = \sqrt{Q}R$  and  $Q$  fixed in a chart, the exact conversions are

$$M_e = Q^{e/2-2A} M_e^*, \quad M_e^* = \int R^e E_e^* dR, \quad b_e^*(R) = Q^{(e+1)/2} b_e(\sqrt{Q}R), \\ \Sigma_e(R) = Q^{2A} \sigma_e(\sqrt{Q}R), \quad (\partial_R + e/R) \Sigma_e = -E_e^* + b_e^* M_e^*. \quad (6)$$

Substitute  $r = \sqrt{Q}R$  including its measure in the moment, and differentiate  $\Sigma_e$  to prove each factor. The input  $F_e$  is already auxiliary independent. Taking only an averaged moment of an unaveraged  $F(R, Y)$  would not suffice:  $F = b_e(R) \cos(2\pi Y_1)$  has zero averaged moment but a nonzero exterior primitive at each generic  $Y$ .

## 4 The source covariance derivative and its signed right inverse

At each slow point let  $w_{\text{tan}} = (w_\theta, w_z)$  and define

$$\mathcal{C}(w) = \langle w_r w_{\text{tan}} \rangle, \quad \mathcal{B}(w, v) = \langle w_r v_{\text{tan}} + v_r w_{\text{tan}} \rangle.$$

Direct expansion proves  $\mathcal{C}(w + v) = \mathcal{C}(w) + \mathcal{B}(w, v) + \mathcal{C}(v)$  and  $D\mathcal{C}(w)[v] = \mathcal{B}(w, v)$ .

Use the exact source real pulses  $b_\sigma = \chi_g(\xi_g)\psi(v)t_\sigma^h \cos(k\Phi_\sigma)$ ,  $\sigma \in \{+, -\}$ . They exclude the slow cutoff  $\eta_\beta$  and scalar amplitudes;  $\chi_g$  is an auxiliary transverse cutoff. The two signs have disjoint auxiliary supports. Put  $H_{\text{cov}} = [\mathcal{C}(b_+) \mid \mathcal{C}(b_-)]$ . For slow scalar coefficients independent of both averages,

$$\mathcal{C}(a_+ b_+ + a_- b_-) = H_{\text{cov}}(a_+^2, a_-^2)^T.$$

All cross products vanish by disjointness. The factor  $1/2$  from  $\langle \cos^2(k\Phi_\sigma) \rangle_\theta$  is already inside the columns.

On the source interior cone the fixed primary amplitudes satisfy  $a_\sigma = \sqrt{(H_{\text{cov}}^{-1} T_{0,*})_\sigma} > 0$ ,  $T_{0,*} = (Q/q)^{A+1/2} T_0$ , and  $W_0 = \sqrt{\varepsilon} \sum_\sigma a_\sigma b_\sigma$ . For any signed slow real vector  $\Sigma$ , define

$$d_\Sigma = H_{\text{cov}}^{-1} \Sigma / \varepsilon, \quad \delta a_\sigma = (d_\Sigma)_\sigma / (2a_\sigma), \quad L_\Sigma = \sqrt{\varepsilon} \sum_\sigma \delta a_\sigma b_\sigma. \quad (7)$$

It follows exactly that

$$\mathcal{B}(W_0, L_\Sigma) = \varepsilon H_{\text{cov}}(2a \odot \delta a) = \Sigma, \quad (8)$$

$$\mathcal{C}(W_0 + L_\Sigma) = \varepsilon T_{0,*} + \Sigma + \varepsilon H_{\text{cov}}(\delta a_+^2, \delta a_-^2)^T. \quad (9)$$

This proves a linear right inverse of the derivative. It is not an exact inverse of the quadratic covariance map. There is no sign or size restriction on  $\Sigma$  in these identities; no square root of  $H_{\text{cov}}^{-1}(T_{0,*} + \Sigma/\varepsilon)$  is taken.

With Euclidean and induced operator norms and  $a_{\min} = \min_\sigma |a_\sigma| > 0$ , they give the explicit bound

$$\|\mathcal{C}(L_\Sigma)\| \leq \frac{\|H_{\text{cov}}\| \|H_{\text{cov}}^{-1}\|^2}{4\varepsilon a_{\min}^2} \|\Sigma\|^2. \quad (10)$$

Indeed  $\|\delta a\| \leq \|H_{\text{cov}}^{-1}\| \|\Sigma\| / (2\varepsilon a_{\min})$  and  $\|(\delta a_\sigma^2)_\sigma\|_2 \leq \|\delta a\|_2^2$ . All higher derivative costs are explicit from Leibniz and

$$\partial^I H_{\text{cov}}^{-1} = -H_{\text{cov}}^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J H_{\text{cov}}) \partial^{I-J} H_{\text{cov}}^{-1},$$

and, for any nonzero scalar  $a$ ,  $\partial^I(a^{-1}) = -a^{-1} \sum_{0 < J \leq I} \binom{I}{J} (\partial^J a) \partial^{I-J}(a^{-1})$ . Together with (7) these recurrences give finite bounds for each order without assuming constant amplitudes or constant columns.

## 5 Assembly, the actual covariance change, and residual cancellation

For the physical target  $\sigma = (\sigma_2, \sigma_1)$  from (4), use the source squared slow partition  $\sum_\beta \eta_\beta^2 = 1$  on its support. Set

$$\begin{aligned} W_{0,\text{phys}} &= \sum_\beta Q_\beta^{-A} \eta_\beta W_{0,\beta}, \\ v_{\text{pr}} &= \sum_\beta Q_\beta^{-A} \eta_\beta L_\beta (Q_\beta^{2A} \sigma). \end{aligned} \quad (11)$$

Overlapping slow labels have disjoint auxiliary supports. Taking cross covariance and using (8) gives

$$\mathcal{B}(W_{0,\text{phys}}, v_{\text{pr}}) = \sum_{\beta} Q_{\beta}^{-2A} \eta_{\beta}^2 Q_{\beta}^{2A} \sigma = \sigma.$$

Every band scale remains distinct. Likewise the leading physical factor is  $Q^{-2A} \varepsilon(Q/q)^{A+1/2} = Q^{-A+h+1/2} q^{-A-1/2} = q^{-A-1/2}$  because  $A = 1/2 + h$ .

Apply the exact curl (1) to each coefficient of  $v_{\text{pr}}$ , including the slow cutoffs before curling. Write the resulting divergence-free increment as  $v_s = v_{\text{pr}} + r_s$ . Let  $t_{s,m}$  denote the chart transverse harmonic coefficients of  $\eta_{\beta} L_{\beta}(Q_{\beta}^{2A} \sigma)$ , before multiplication by  $Q_{\beta}^{-A}$ . For each such coefficient choose  $\pi_{s,m} = i(n^T K t_{s,m} - (n')^T t_{s,m}) / (k_{\beta} m |n|^2)$  and sum  $\text{Re}(Q_{\beta}^{-2A} \pi_{s,m} e^{i\phi_{\beta,m}})$  to define the smooth compact  $\pi_s$ . Its full gradients remain below. Let the actual old wave  $w$  consist of the wave part of  $u_*$  plus the candidate  $w_y$ , including all existing curl corrections, and write  $w = W_{0,\text{phys}} + e_w$ . Define tensor covariance  $\mathcal{T}(w) = \langle w \otimes w \rangle$  and  $\mathcal{B}_T(u, v) = \langle u \otimes v + v \otimes u \rangle$ . Expansion, with  $e_w = w - W_{0,\text{phys}}$ , proves

$$\Delta \mathcal{T} = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}}) + \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s). \quad (12)$$

Its first radial–tangential pair is  $\sigma$ . All other entries and the remaining three tensors are retained. For example their radial–tangential pair has bound

$$2\|e_w\|_{L_{\theta,Y}^2} \|v_{\text{pr}}\|_{L_{\theta,Y}^2} + 2\|w\|_{L_{\theta,Y}^2} \|r_s\|_{L_{\theta,Y}^2} + \|v_s\|_{L_{\theta,Y}^2}^2,$$

by Cauchy–Schwarz. The compact-set estimates above and the explicit curl differentiate to bound all these terms at each finite order.

For clarity the complete residual after adding  $v_s$  and its selected pressure  $\pi_s$  is, without any averaging,

$$\begin{aligned} R_{\nu_{\text{NS}}}(u + v_s, p + \pi_s) &= R_{\nu_{\text{NS}}}(u, p) + \partial_t v_s + (u \cdot \nabla) v_s + (v_s \cdot \nabla) u \\ &\quad - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s + (v_s \cdot \nabla) v_s. \end{aligned} \quad (13)$$

This follows by expanding the product and retains viscosity and every pressure term. For a symmetric tensor  $S$  the averaged tangential divergence in physical cylindrical coordinates is

$$(\langle \nabla \cdot S \rangle)_{\theta} = (\partial_r + 2/r) \langle S_{r\theta} \rangle + \partial_z \langle S_{z\theta} \rangle, \quad (\langle \nabla \cdot S \rangle)_z = (\partial_r + 1/r) \langle S_{rz} \rangle + \partial_z \langle S_{zz} \rangle.$$

To verify, write the Cartesian divergence in the orthonormal cylindrical basis, using  $\partial_{\theta} e_r = e_{\theta}$ ,  $\partial_{\theta} e_{\theta} = -e_r$ ; periodic angular derivatives average to zero and the two basis terms give  $2/r$  and  $1/r$  respectively. The auxiliary chain terms average to zero by periodicity, but the physical slow derivatives remain.

Let  $U = u_* + w_y$ ,  $P_y = p_* + \pi_y$ , and set

$$\mathcal{L} = \partial_t v_s + (U \cdot \nabla) v_s + (v_s \cdot \nabla) U - \nu_{\text{NS}} \Delta v_s + \nabla \pi_s.$$

For a fully explicit averaged remainder, write  $U = u_{\text{mean}} + w$ , where  $u_{\text{mean}}$  is its angular mean, and put  $\mathcal{V} = \mathcal{L} - (w \cdot \nabla) v_s - (v_s \cdot \nabla) w$ . Then the average of (13) is  $\langle R_y \rangle + \langle \mathcal{V} \rangle + \nabla \cdot \Delta \mathcal{T}$ . Let  $S_0 = \mathcal{B}_T(W_{0,\text{phys}}, v_{\text{pr}})$  and  $S_{\text{rem}} = \mathcal{B}_T(e_w, v_{\text{pr}}) + \mathcal{B}_T(w, r_s) + \mathcal{T}(v_s)$ . Consequently the radial divergence of the first pair in (12) changes the prescribed averaged residual  $F_e$  into  $b_e M_e$  by (5).

The axial tensor divergences, the other three tensors, the exact linear terms and the pressure terms of (13) remain in the new residual. Equivalently the new averaged tangential residual is exactly

$$\begin{aligned}\langle R_{\text{new}} \rangle_{\text{tan}} &= \langle R_{\nu_{\text{NS}}}(u_*, p_*) \rangle_{\text{tan}} + (b_2 M_2, b_1 M_1) + \langle \mathcal{V} \rangle_{\text{tan}} \\ &+ \partial_z((S_0)_{z\theta}, (S_0)_{zz}) + (\nabla \cdot S_{\text{rem}})_{\text{tan}}.\end{aligned}$$

Every term here is defined by the displayed differentiated fields and tensors. None is assumed to vanish. The map from  $y$  through (3), (4), (7), and (11) is now an actual composite construction. It retains the signed residual produced by the coupled candidate and the source's original physical and chart units.

## 6 Signs and the exact full-flow reflection

The maps  $F_e \mapsto M_e \mapsto \sigma_e \mapsto L_\Sigma$  are linear and reverse sign with their inputs;  $\mathcal{C}(L_\Sigma)$  is even. The two primary amplitudes may also be replaced independently by  $s_\sigma a_\sigma$ ,  $s_\sigma \in \{-1, 1\}$ , provided the same fixed choice is used in each denominator: the cross identity is unchanged. This does not negate the nonlinear Navier–Stokes equation.

Here is the exact full-flow map. In Cartesian coordinates let  $O = \text{diag}(1, -1, 1)$  and define

$$u^\sharp(x, t) = Ou(Ox, t), \quad p^\sharp(x, t) = p(Ox, t), \quad f^\sharp(x, t) = Of(Ox, t).$$

The chain rule gives  $\nabla u^\sharp = O(\nabla u)(Ox, t)O$ ,  $\Delta u^\sharp = O(\Delta u)(Ox, t)$ , and  $(u^\sharp \cdot \nabla)u^\sharp = O((u \cdot \nabla)u)(Ox, t)$ . Thus  $\nabla \cdot u^\sharp = 0$  and  $R_{\nu_{\text{NS}}}(u^\sharp, p^\sharp) = OR_{\nu_{\text{NS}}}(u, p)(Ox, t)$  with the same positive viscosity. In cylindrical components this is

$$(u_r^\sharp, u_\theta^\sharp, u_z^\sharp)(r, \theta, z, t) = (u_r, -u_\theta, u_z)(r, -\theta, z, t),$$

and the force transforms identically. Pressure reflects in its argument. For an axisymmetric field this reverses exactly its swirl; meridional components remain fixed. The entire momentum tensor is mapped to  $OTO^T$ , so its radial–tangential pair transforms by  $(T_{r\theta}, T_{rz}) \mapsto (-T_{r\theta}, T_{rz})$  after angular averaging. Vorticity transforms as  $\omega^\sharp(x, t) = \det(O)O\omega(Ox, t)$ , obtained by applying the same derivative identities to the Levi-Civita formula for curl. The map is an involution, preserves the kinetic-energy integral (by  $|\det O| = 1$ ), compact support, smoothness, and blowup magnitude along the reflected path. It preserves zero initial data when present. It proves transport of any established axisymmetric signed swirl divergence to the opposite sign; it does not itself prove an endpoint.

The source five-moment correction, auxiliary mean inverse, and infinite summation are subsequent source operations. Their full estimates are not re-proved here. The earlier draft formulas for them are withdrawn: in particular the fifth moment row is  $\int (2RG\gamma_d - RV\Delta v) \, dR = -J_z$ , without an extra outer  $R$ , and its reserved profile in a  $Q$  chart includes  $(Q/q)^A$ . The finite map proved above retains the two moments and all other residual terms, so it establishes neither a third shooting return nor an infinite modified viscous sequence.